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Vellore Institute of Technology
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School of Computer Science Engineering and Information Systems
(SCORE)

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Digital Assignment – I

Course name: Software Engineering Economics

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22MIS0072

Analyse a real-world or proposed project by applying the concepts of the Project Life Cycle (PLC). Break down the project into the five phases of the PLC and evaluate key challenges, strategies, and outcomes in each phase.

Chosen Project

Development of an “**untitled video game**” from **GameBlip Studios**.
(hypothetical)

Scope: Details about our Untitled Video game

The running draft title for our game is “**K-03 A-20: A KAtchy Journey.**”

The scope of our project is to develop a 2.5D platformer game with rich storytelling, compelling side characters, and a mix of combat mechanics like attacking, shooting, blocking, and mainly parrying.

Players can upgrade their characters through shops, enhancing their abilities and gameplay experience.

Purpose:

To create a fun, catchy, weekend gaming experience with simple controls but deep, heartbreaking narrative with compelling combat mechanics.

Unique Selling point: A great unique selling point (USP) for our game would be, the gameplay and storytelling are an inspiration from popular award-winning titles, “**Hollow Knight**” and “**Sekiro: Shadows die twice.**”

Resources:

- i) **Human Resources:** Game designers, Game developers, Sound designers, Artists and QA testers.
- ii) **Tools for Development:**
 - Game Engine: Unity engine.
 - Art tools: Blender for 2D/3D asset creation and character modelling.
 - Audio Tools: Audacity for sound mixing and music composition.
 - Project Management: Taiga for task management and collaboration among team members.

iii) Budget:

Total budget: **\$53,000 - \$70,000**

Development cost: \$10,000 - \$15,000

Testing and QA: \$5,000 - \$7,000

Marketing and Promotion: \$25,000 - \$30,000

Launching and Initial maintenance: \$5000 - \$10,000

Contingency: \$4,000 - \$8,000

Project Life Cycle:

Now that the project scope, purpose, and resources are discussed, we are going to apply the concepts of Project Life Cycle (PLC) and we are going to breakdown the project to evaluate its key challenges, strategies, and outcomes in each phase.



Figure: Five Phases of PLC

Phase 1: INITIATION

- ➔ The main activities of this phase are to define the project objectives and scope.
- As mentioned before, the project scope is to develop a 2.5D platformer game with rich storytelling, compelling side characters, and a mix of combat mechanics like attacking, shooting, blocking, and mainly parrying.
 - The conduction of feasibility analysis and market research is also done in this phase.
 - For our project, the gaming market is at its all-time peak now. As of 2023, the gaming industry was projected to generate over **\$365 billion** in revenue. With a great narrative and fun combat mechanics, our game can produce a fortune!

- ➔ Some of the challenges for this would be securing funding and resources from our investors and developers. We have to make sure that the game concept is unique and marketable with promotions and word of mouth.
- ➔ A great Strategy would be to pitch a clear game concept (As mentioned in the scope of the project) and refine it with brainstorming and visual storyboarding.
- ➔ The Outcome from this phase would be project approval and initial funding secured. All the required documents are initiated and recorded properly.

Phase 2: PLANNING

- ➔ The main activities of this stage would be to create a detailed plan with milestones. Also to allocate resources and setting a budget for each department. Lastly to discuss the risk management strategies to avoid any big failures during the development cycle of the game.
- ➔ The main challenge will be to estimate time and resource with accuracy. The game cannot stay in development for a long time, people will lose interest and they will lose faith in our game. So, to estimate the development time and required resources is important with coordinating across a multidisciplinary team.
- ➔ Key Strategy is to use project management tool to keep track of the development of each part of the game (Ex: attacking mechanism, interaction mechanism, parrying mechanism, etc.) like Taiga. Also to conduct regular team meetings for progress tracking.
- ➔ The Outcome from this stage would be a comprehensive roadmap for game development.

Phase 3: EXECUTION

- ➔ After getting the roadmap, it is time to develop the game. The activities performed will be developing game mechanics (movement, combat, upgrades, UI/UX), Designing game levels, character models and ambient environments. Main part is to write the script of the story and implement the story and dialogues for each character.
- ➔ The Challenging part of this phase is to stick to the schedule of the development and to not go overboard with our budget. With the game, it would be to balance quality with development speed, ensuring that the game mechanics correlate with the story and are engaging and bug-free.
- ➔ The Strategy to achieve that would be to follow an iterative development process. Agile development model would be perfect for game development.

- ➔ Outcome will be a fully developed game that is ready for testing.

Phase 4: MONITORING AND CONTROL

- ➔ The key activity of this phase would be to track progress against the project plan, whether we are sticking with the schedule or not. Also to test the game for any bugs, inconvenient experiences and QA testing. This testing will help our development team to address bugs and to optimize performance.
- ➔ The main challenge will be to manage scope creeps. For our project, it will be like adding features, adding new mechanics or even adding few story plots. We have to conduct a thorough analysis before thinking about adding them to the completed product.
- ➔ Strategy for this phase's success would be to prioritize tasks based on impact and deadlines. Also use version control tools to keep track of the progress like GitHub.
- ➔ Outcome will be a stable, fun, and envisioned game that is ready for release.

Phase 5: CLOSURE

- ➔ Getting with Steam and signing a contract to release our game is the main activity of this phase. After release, get feedback from players and evaluate our game's performance.
- ➔ The main Challenges would be to maintain the game's stability post-launch and to fix any bugs and updates. Also measuring success against KPIs (downloads, reviews, revenue)
- ➔ Offering post-launch support and updates, collaborating with social media influencers who are passionate and enthusiastic about indie game development would be a great strategy to increase the game's popularity.
- ➔ Outcome will be a successful game release with plans for future updates.

Critical Evaluation

- Timeliness: We have to analyse if the milestones were met as planned or not. This will affect the project's development and stakeholder's interest.
- Budget: Regular reviews to see if the project stayed within budget or not.
- Scope: We need to assess whether all planned features were implemented or not. This will be a huge problem if the envisioned game did not meet the game that is being made.
- Challenges: The development manager needs to identify areas for improvement. (Ex: delays in execution, unanticipated bugs, etc.)

Recommendations for Future Projects

The studio will be in a good place after the release of the first title. For future plans, we can use advanced project management tools for better tracking.

Also to include great people with unique ideas and creativity to create all new games with interesting plot and lore.

To include a contingency buffer in the budget and timeline.

To conduct extensive pre-launch testing to reduce post-launch issues.

Also, we can conduct game demo sessions at big gaming conventions for players to experience our game at the early stages and give feedbacks for better improvement and ideas.

Conclusion

In conclusion, the development of “K-03 A-20: A KAtchy Journey” is showing the importance of a narrow approach through the project life cycle (PLC). The project's overall success was greatly influenced by each phase.

Initiation created a clear vision, planning made sure resources were allocated and risks were removed, execution made the concept a reality through iterative development, monitoring ensured quality and timeline adherence, and closure produced a finished product for the market. The team's resilience was put to the test by issues like scope creep and post-launch updates, but they were successfully handled by using proactive tactics. These revelations highlight the value of flexibility and ongoing learning in project management, offering a solid basis for enhancing upcoming initiatives.

References:

[What Is the Project Life Cycle? - GeeksforGeeks](#)

[Video Game Market Size, Share And Growth Report, 2030](#)

[Nine Sols on Steam](#)

[20 Best 2.5D Metroidvania Games That You Should Play ! - YouTube](#)